# 04 - Allen-Cahn Equation Experiments

**AC-PINN Project** | Authors: Suyash Vasal Jain, Nishita Raghvendra

**PDE:** $u_t = \varepsilon^2 u_{xx} + u - u^3$, $\varepsilon = 0.01$

**IC:** $u(x,0) = x^2 \cos(\pi x)$ | **BC:** $u(-1,t) = u(1,t) = 0$

**Domain:** $x \in [-1,1]$, $t \in [0,1]$

**Architecture:** `[2, 128, 128, 128, 128, 128, 1]` | **Epochs:** 10000

**Note:** Allen-Cahn is the hardest test. Stiff PDE with sharp interface. Vanilla PINN is known to fail here. This is AC-PINN's strongest showcase.

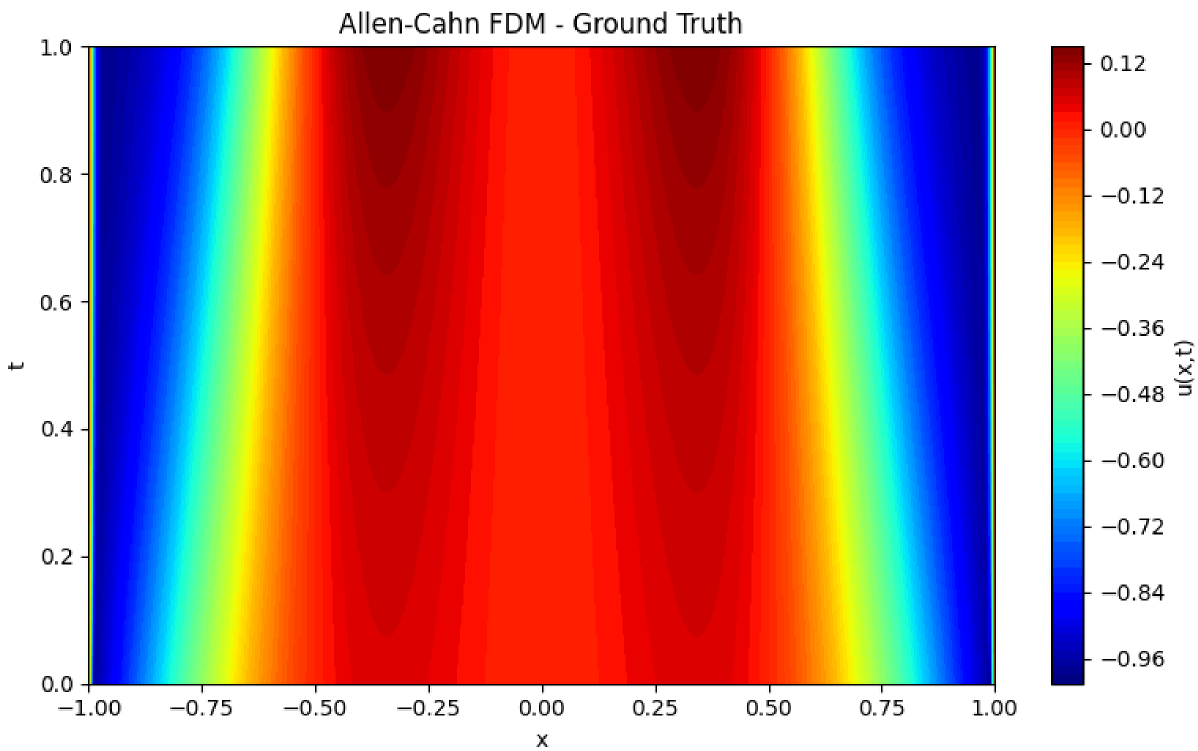
In [1]:
```
import sys, os
sys.path.append('..')
import torch
import numpy as np
import matplotlib.pyplot as plt
from pinn_base import (
    device, NoisyDataGenerator, PINNSolver, ACPINNSolver,
    AllenCahnFDM, Benchmark, save_metrics, save_history, save_training_plots
)

PDE        = 'allen_cahn'
LAYERS     = [2, 128, 128, 128, 128, 128, 1]
EPOCHS     = 10000
PDE_PARAMS = {'epsilon': 0.01}
RESULTS    = '../results/allen_cahn/'
FIGURES    = '../figures/allen_cahn/'
os.makedirs(RESULTS, exist_ok=True)
os.makedirs(FIGURES, exist_ok=True)

gen = NoisyDataGenerator(pde=PDE)
print(f'Device: {device}')
```

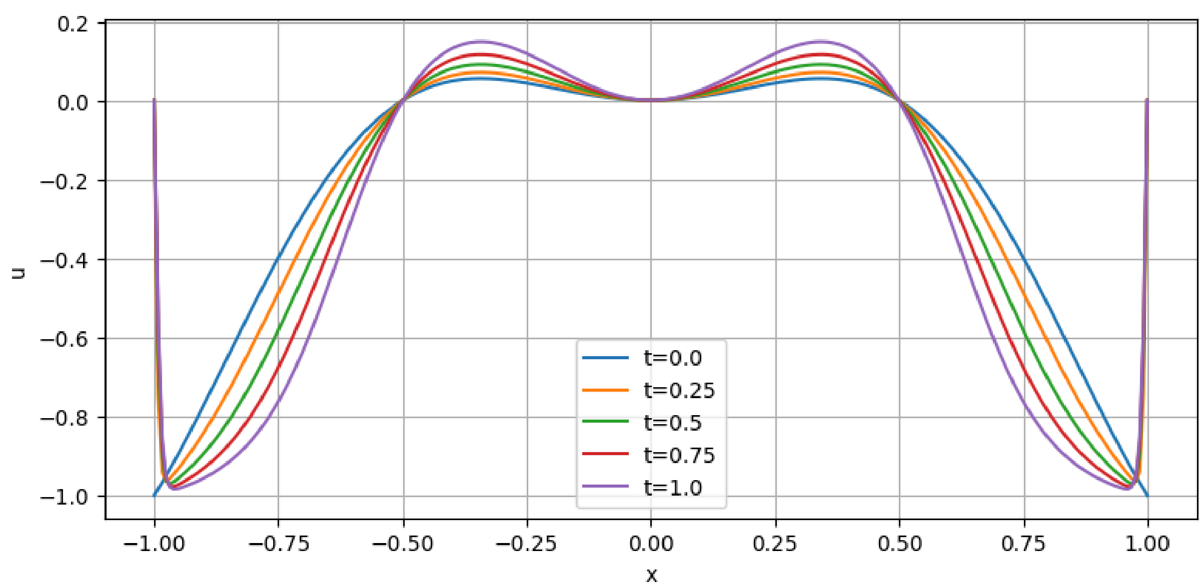
```
Device: cuda
```

# Section 1 - FDM Ground Truth

In [2]:
```
fdm = AllenCahnFDM(nx=256, nt=5000, epsilon=0.01)
fdm.solve()
fdm.plot_solution(title='Allen-Cahn FDM - Ground Truth')
fdm.plot_time_slices()
```

```
AllenCahnFDM solved in 1.9181s
```




# Section 2 - Data Conditions

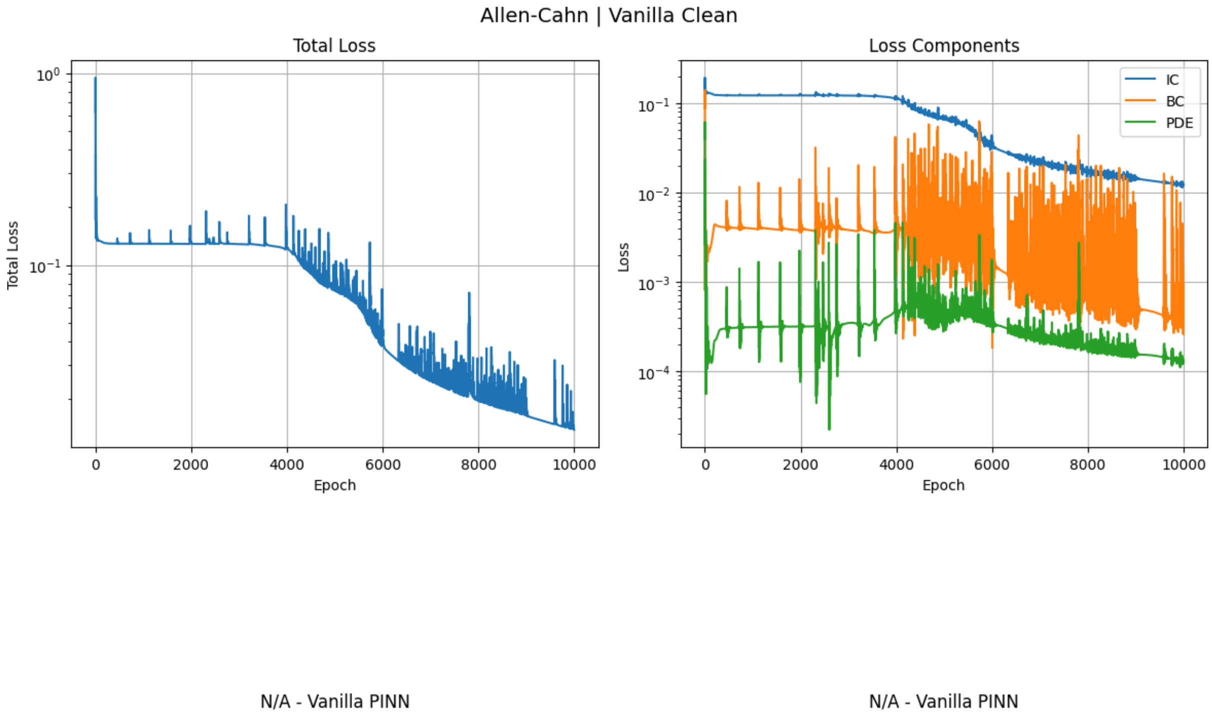
In [3]:
```
# Allen-Cahn needs more collocation points due to stiffness
data_clean_dense  = gen.generate(N_ic=1000, N_bc=1000, N_f=10000, noise_eps=0.0)
data_noisy_sparse = gen.generate(N_ic=20,   N_bc=20,   N_f=3000,  noise_eps=0.1)
print('Data ready')
```

```
Data ready
```

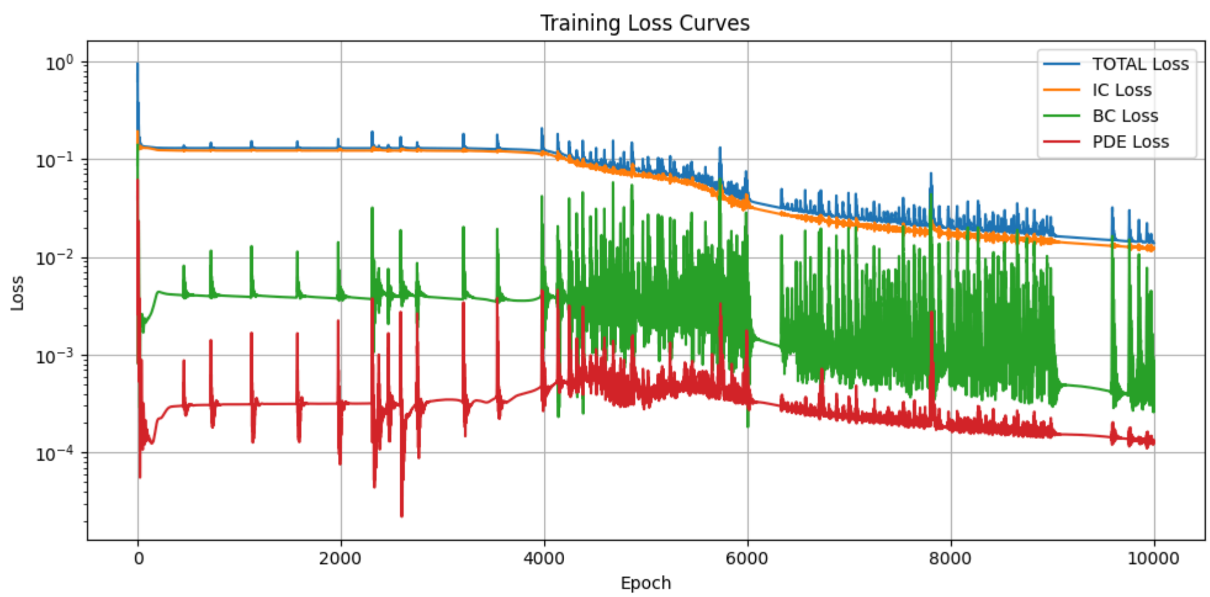
# Section 3 - Vanilla PINN, Clean Dense

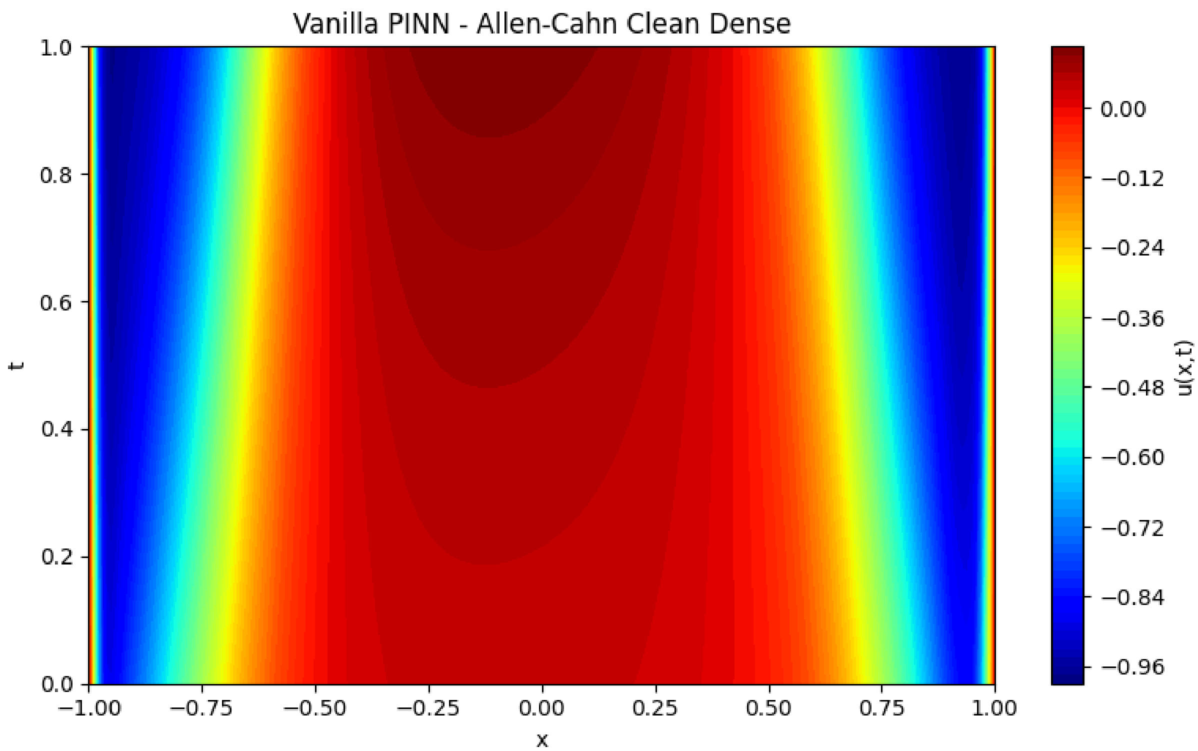
In [4]:
```
# Higher lambda_pde for stiff PDE
vanilla_clean = PINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS,
                           lambda_ic=1.0, lambda_bc=1.0, lambda_pde=10.0)
h_vc = vanilla_clean.fit(data_clean_dense, epochs=EPOCHS, print_every=1000, label='
save_training_plots(h_vc, FIGURES+'vanilla_clean_training.png', 'Allen-Cahn | Vanil
vanilla_clean.plot_loss_history(h_vc)
vanilla_clean.plot_solution(title='Vanilla PINN - Allen-Cahn Clean Dense')
vanilla_clean.plot_initial_condition_comparison(gen)
save_history(h_vc, RESULTS+'vanilla_clean_history.npy')
```

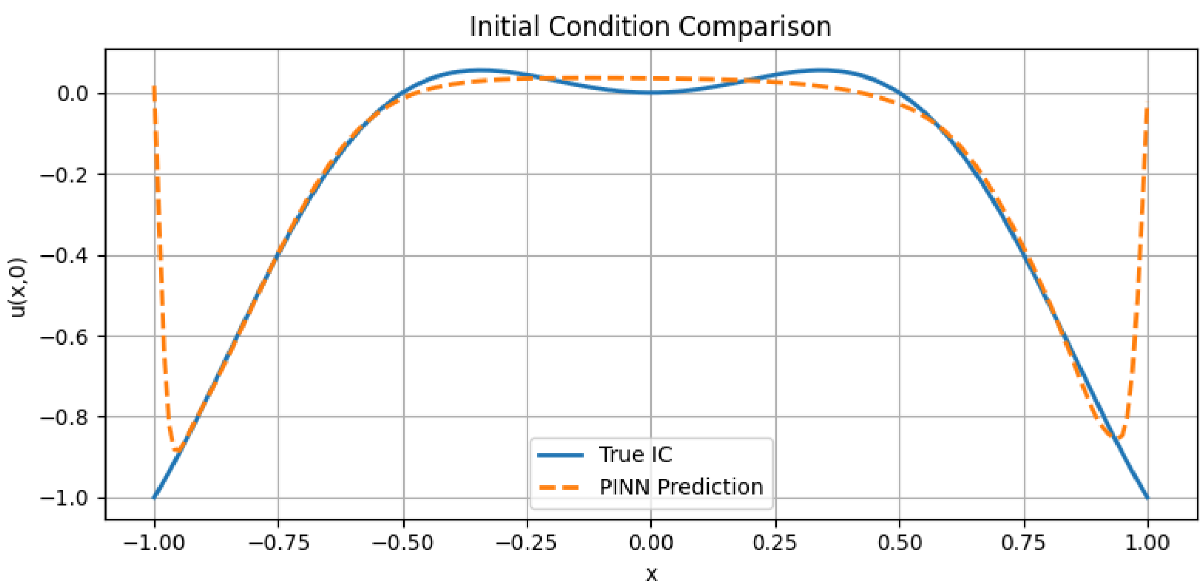
```
D:\PINN\ac-pinn-project\ac-pinn-project\venv\Lib\site-packages\torch\autograd\graph.
py:825: UserWarning: Attempting to run cuBLAS, but there was no current CUDA contex
t! Attempting to set the primary context... (Triggered internally at C:\actions-runn
er\_work\pytorch\pytorch\builder\windows\pytorch\aten\src\ATen\cuda\CublasHandlePoo
l.cpp:135.)
  return Variable._execution_engine.run_backward(  # Calls into the C++ engine to ru
n the backward pass
[Allen-Cahn | Vanilla Clean | Epoch     0/10000] Total: 0.626974 | IC: 0.144497 | B
C: 0.087670 | PDE: 0.039481
[Allen-Cahn | Vanilla Clean | Epoch  1000/10000] Total: 0.128943 | IC: 0.121899 | B
C: 0.003938 | PDE: 0.000311
[Allen-Cahn | Vanilla Clean | Epoch  2000/10000] Total: 0.130245 | IC: 0.124084 | B
C: 0.004663 | PDE: 0.000150
[Allen-Cahn | Vanilla Clean | Epoch  3000/10000] Total: 0.128454 | IC: 0.121195 | B
C: 0.003796 | PDE: 0.000346
[Allen-Cahn | Vanilla Clean | Epoch  4000/10000] Total: 0.124353 | IC: 0.117401 | B
C: 0.001691 | PDE: 0.000526
[Allen-Cahn | Vanilla Clean | Epoch  5000/10000] Total: 0.074803 | IC: 0.068328 | B
C: 0.002227 | PDE: 0.000425
[Allen-Cahn | Vanilla Clean | Epoch  6000/10000] Total: 0.048437 | IC: 0.040147 | B
C: 0.003748 | PDE: 0.000454
[Allen-Cahn | Vanilla Clean | Epoch  7000/10000] Total: 0.034409 | IC: 0.023831 | B
C: 0.006994 | PDE: 0.000358
[Allen-Cahn | Vanilla Clean | Epoch  8000/10000] Total: 0.019724 | IC: 0.017373 | B
C: 0.000474 | PDE: 0.000188
[Allen-Cahn | Vanilla Clean | Epoch  9000/10000] Total: 0.020294 | IC: 0.015079 | B
C: 0.003662 | PDE: 0.000155
Training complete in 276.59s
```




```
Saved: ../figures/allen_cahn/vanilla_clean_training.png
Saved: ../figures/allen_cahn/vanilla_clean_training.csv
```





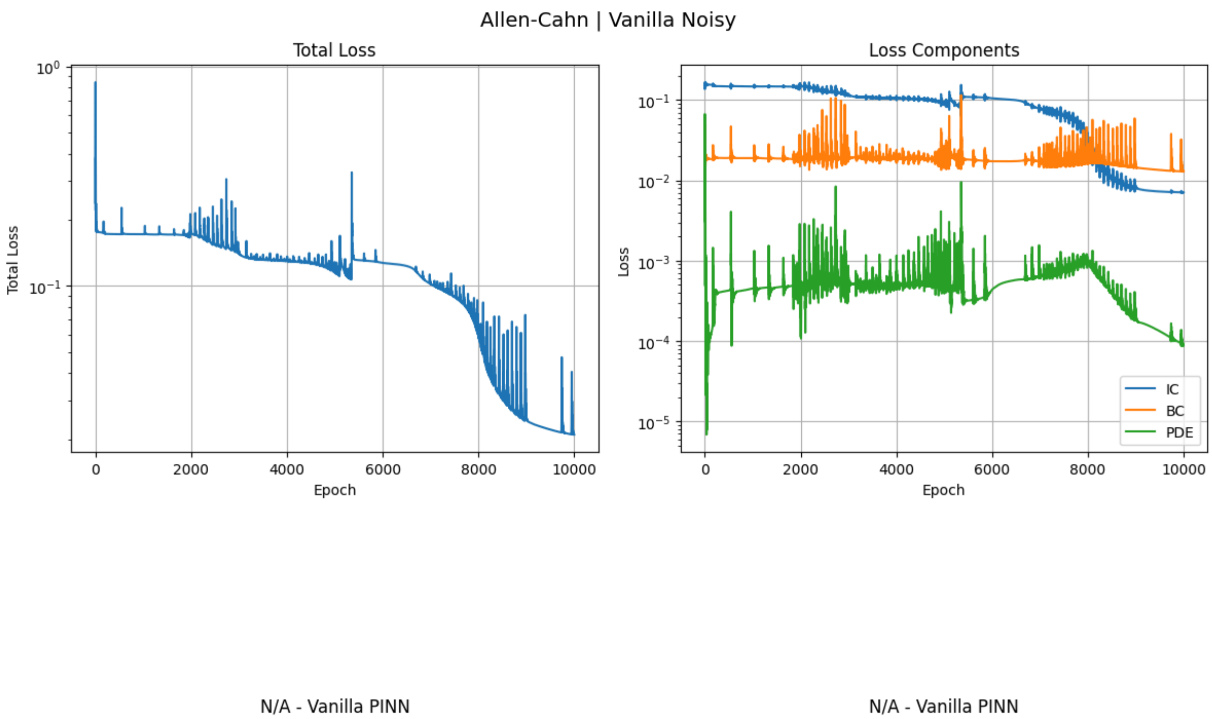

```
Saved: ../results/allen_cahn/vanilla_clean_history.npy
```

# Section 4 - Vanilla PINN, Noisy Sparse

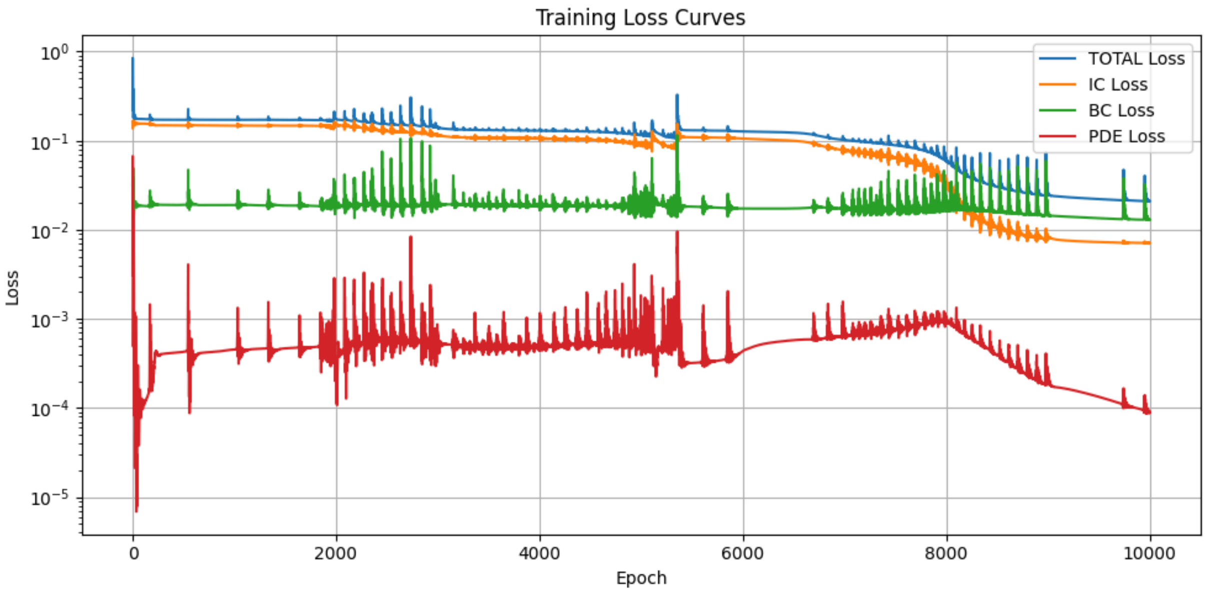
⚠️ Vanilla PINN typically fails here on Allen-Cahn.

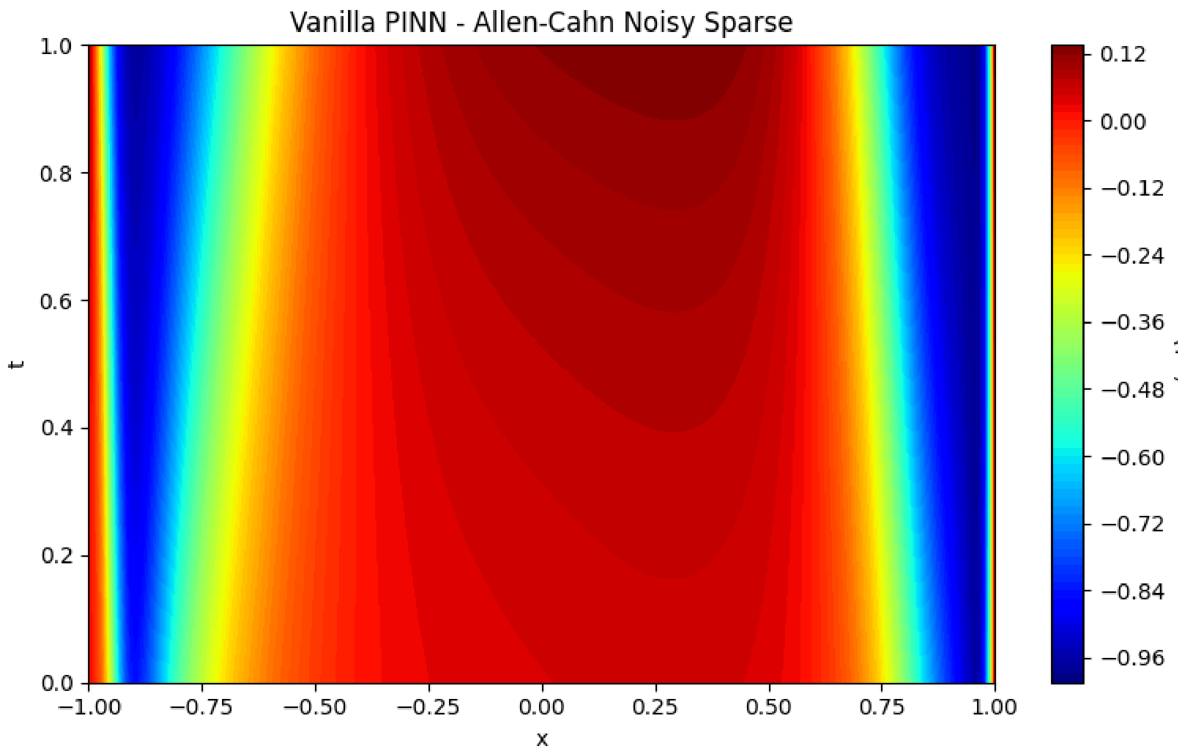
In [5]:
```
vanilla_noisy = PINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS,
                           lambda_ic=1.0, lambda_bc=1.0, lambda_pde=10.0)
h_vn = vanilla_noisy.fit(data_noisy_sparse, epochs=EPOCHS, print_every=1000, label=
save_training_plots(h_vn, FIGURES+'vanilla_noisy_training.png', 'Allen-Cahn | Vanil
vanilla_noisy.plot_loss_history(h_vn)
vanilla_noisy.plot_solution(title='Vanilla PINN - Allen-Cahn Noisy Sparse')
save_history(h_vn, RESULTS+'vanilla_noisy_history.npy')
```

```
[Allen-Cahn | Vanilla Noisy | Epoch     0/10000] Total: 0.222897 | IC: 0.160220 | B
C: 0.031515 | PDE: 0.003116
[Allen-Cahn | Vanilla Noisy | Epoch  1000/10000] Total: 0.171356 | IC: 0.147896 | B
C: 0.018931 | PDE: 0.000453
[Allen-Cahn | Vanilla Noisy | Epoch  2000/10000] Total: 0.175861 | IC: 0.149349 | B
C: 0.023983 | PDE: 0.000253
[Allen-Cahn | Vanilla Noisy | Epoch  3000/10000] Total: 0.143990 | IC: 0.113653 | B
C: 0.025421 | PDE: 0.000492
[Allen-Cahn | Vanilla Noisy | Epoch  4000/10000] Total: 0.130944 | IC: 0.102572 | B
C: 0.020978 | PDE: 0.000739
[Allen-Cahn | Vanilla Noisy | Epoch  5000/10000] Total: 0.120333 | IC: 0.096492 | B
C: 0.017651 | PDE: 0.000619
[Allen-Cahn | Vanilla Noisy | Epoch  6000/10000] Total: 0.127031 | IC: 0.105304 | B
C: 0.017335 | PDE: 0.000439
[Allen-Cahn | Vanilla Noisy | Epoch  7000/10000] Total: 0.102742 | IC: 0.075615 | B
C: 0.019211 | PDE: 0.000792
[Allen-Cahn | Vanilla Noisy | Epoch  8000/10000] Total: 0.059195 | IC: 0.033552 | B
C: 0.016160 | PDE: 0.000948
[Allen-Cahn | Vanilla Noisy | Epoch  9000/10000] Total: 0.025510 | IC: 0.008121 | B
C: 0.014946 | PDE: 0.000244
Training complete in 136.67s
```




```
Saved: ../figures/allen_cahn/vanilla_noisy_training.png
Saved: ../figures/allen_cahn/vanilla_noisy_training.csv
```



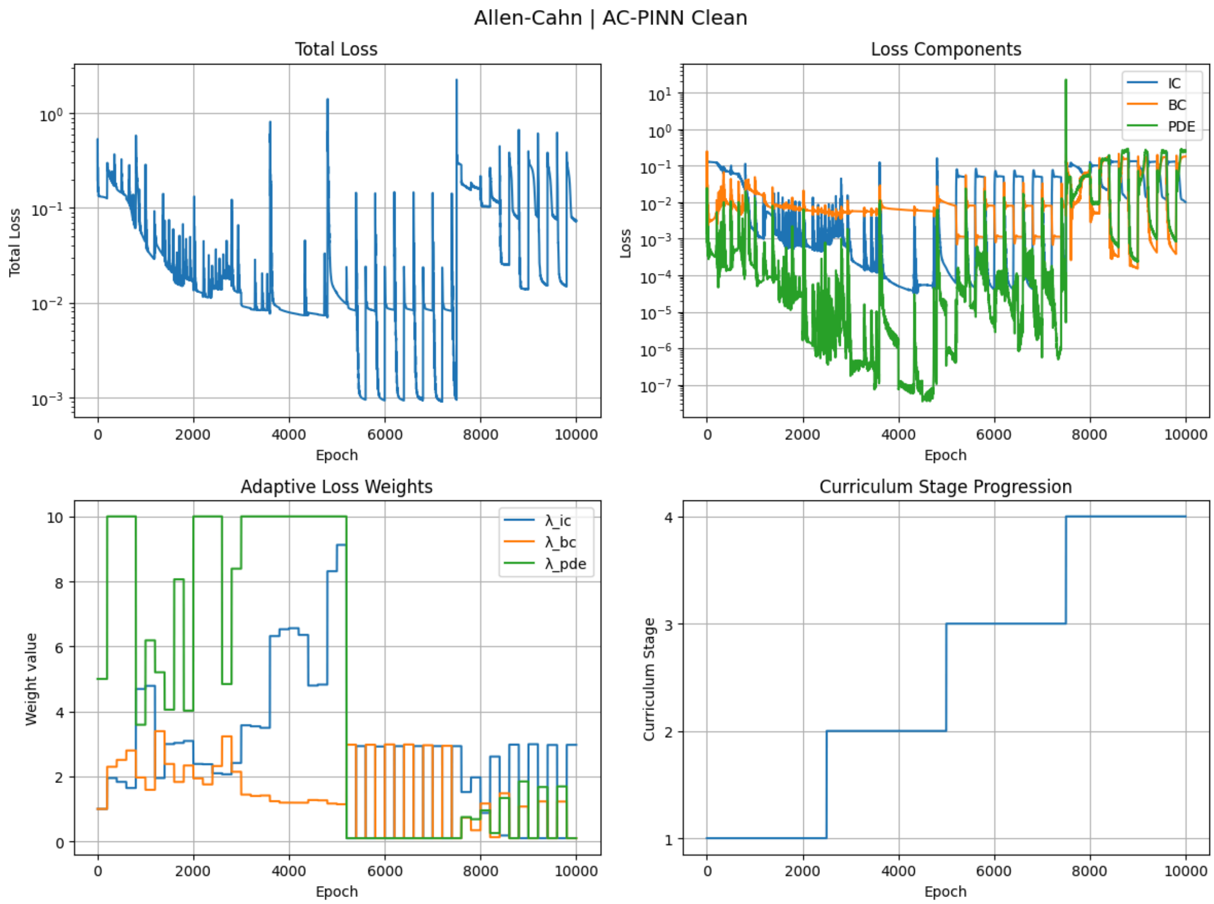

```
Saved: ../results/allen_cahn/vanilla_noisy_history.npy
```

# Section 5 - AC-PINN, Clean Dense

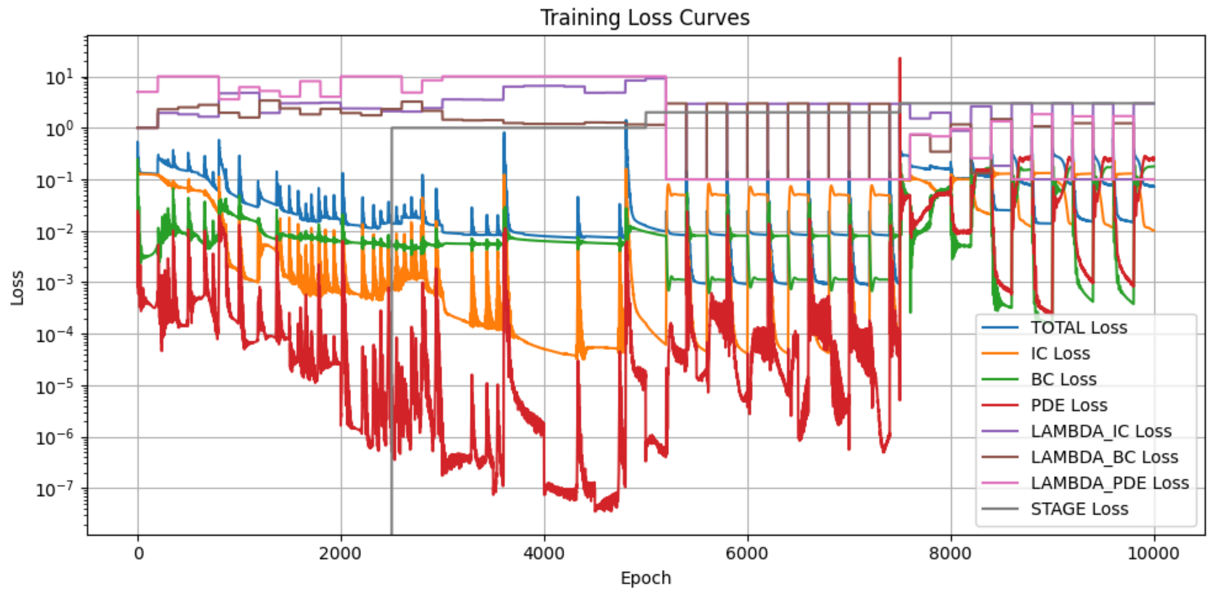
In [6]:
```
ac_clean = ACPINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS,
                        weight_strategy='both',
                        N_pool=30000, resample_every=500)
h_ac = ac_clean.fit(data_clean_dense, epochs=EPOCHS, print_every=1000, label='Allen
save_training_plots(h_ac, FIGURES+'acpinn_clean_training.png', 'Allen-Cahn | AC-PIN
ac_clean.plot_loss_history(h_ac)
ac_clean.plot_weight_history(h_ac)
ac_clean.plot_solution(title='AC-PINN - Allen-Cahn Clean Dense')
```

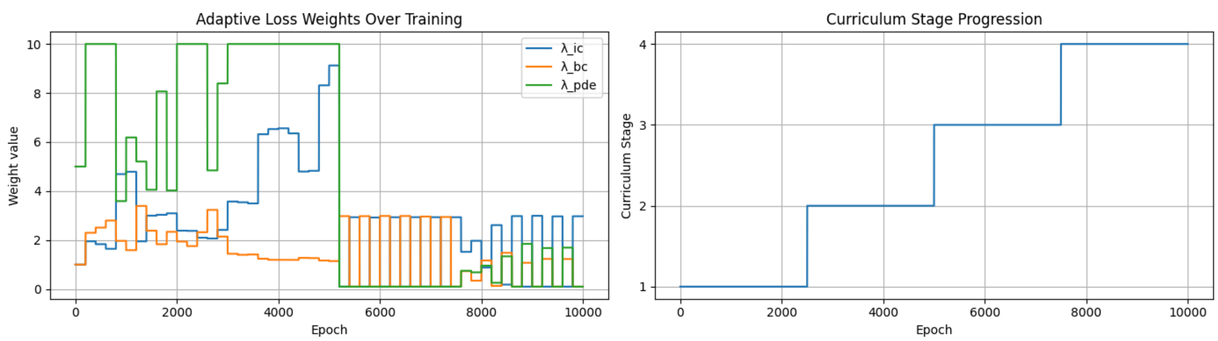
```
ac_clean.plot_initial_condition_comparison(gen)
save_history(h_ac, RESULTS+'ac_clean_history.npy')
```

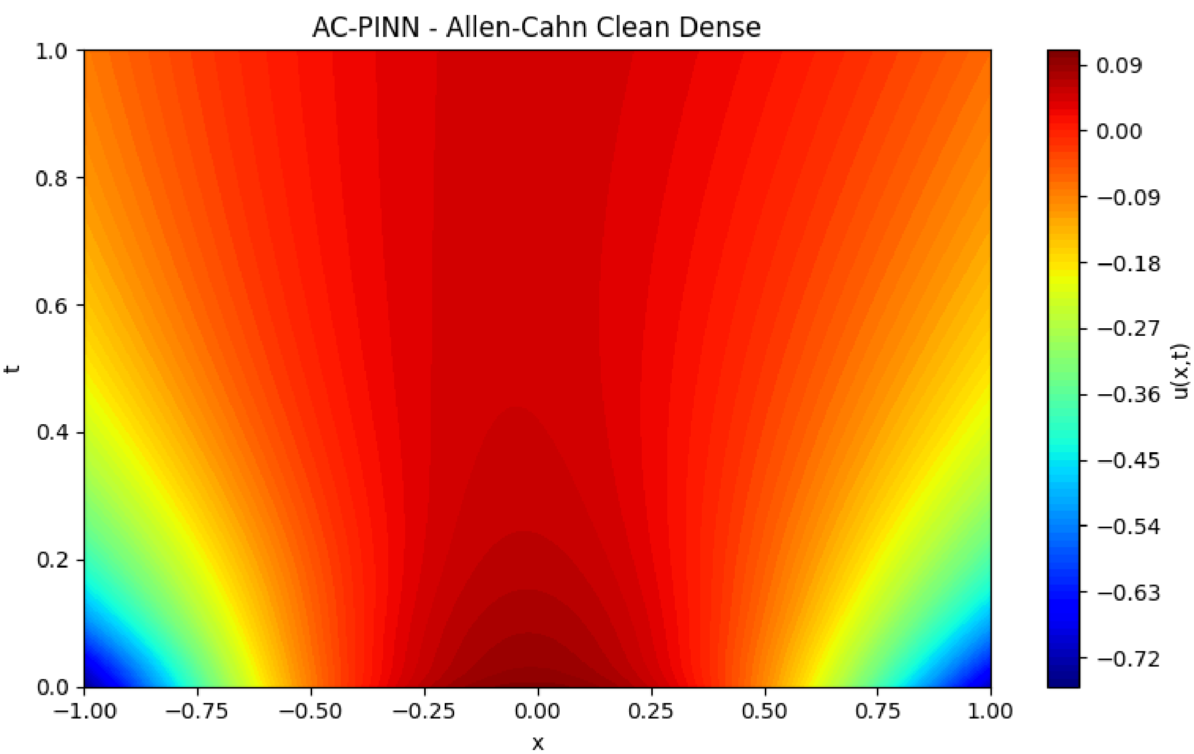
```
[Allen-Cahn | AC-PINN Clean | Epoch     0/10000] Stage 1/4 | Total: 0.196306 | IC: 0.142633 | BC: 0.049597 | PDE: 0.000815 | λ=(1.00,1.00,5.00)
[Allen-Cahn | AC-PINN Clean | Epoch  1000/10000] Stage 1/4 | Total: 0.091380 | IC: 0.006492 | BC: 0.029534 | PDE: 0.002163 | λ=(4.79,1.59,6.19)
[Allen-Cahn | AC-PINN Clean | Epoch  2000/10000] Stage 1/4 | Total: 0.013632 | IC: 0.000618 | BC: 0.006259 | PDE: 0.000002 | λ=(2.38,1.94,10.00)
[Allen-Cahn | AC-PINN Clean | Epoch  3000/10000] Stage 2/4 | Total: 0.009599 | IC: 0.000539 | BC: 0.005325 | PDE: 0.000000 | λ=(3.57,1.44,10.00)
[Allen-Cahn | AC-PINN Clean | Epoch  4000/10000] Stage 2/4 | Total: 0.007876 | IC: 0.000050 | BC: 0.006308 | PDE: 0.000000 | λ=(6.56,1.20,10.00)
[Allen-Cahn | AC-PINN Clean | Epoch  5000/10000] Stage 2/4 | Total: 0.012419 | IC: 0.000160 | BC: 0.009579 | PDE: 0.000000 | λ=(9.12,1.14,10.00)
[Allen-Cahn | AC-PINN Clean | Epoch  6000/10000] Stage 3/4 | Total: 0.023954 | IC: 0.000041 | BC: 0.008028 | PDE: 0.000004 | λ=(0.10,2.98,0.10)
[Allen-Cahn | AC-PINN Clean | Epoch  7000/10000] Stage 3/4 | Total: 0.143955 | IC: 0.049044 | BC: 0.001121 | PDE: 0.000001 | λ=(2.93,0.10,0.10)
[Allen-Cahn | AC-PINN Clean | Epoch  8000/10000] Stage 4/4 | Total: 0.175603 | IC: 0.050622 | BC: 0.067296 | PDE: 0.055103 | λ=(0.88,1.17,0.96)
[Allen-Cahn | AC-PINN Clean | Epoch  9000/10000] Stage 4/4 | Total: 0.397028 | IC: 0.132734 | BC: 0.000153 | PDE: 0.000254 | λ=(2.99,0.10,0.10)
AC-PINN training complete in 299.13s
```

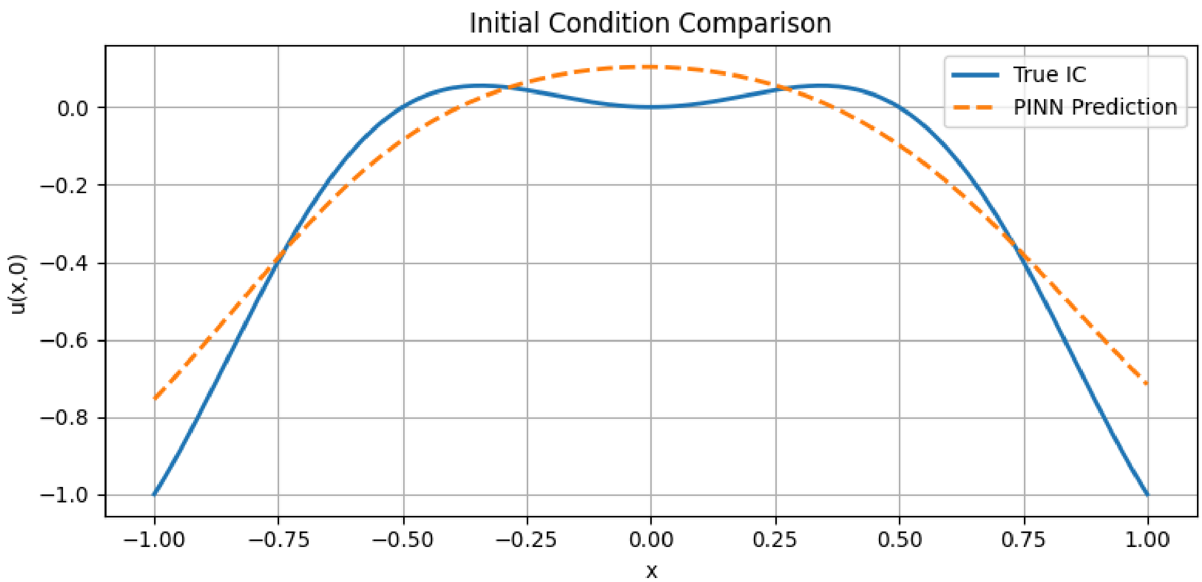



```
Saved: ../figures/allen_cahn/acpinn_clean_training.png
Saved: ../figures/allen_cahn/acpinn_clean_training.csv
```


Training Loss Curves
Loss
Epoch
TOTAL Loss
IC Loss
BC Loss
PDE Loss
LAMBDA_IC Loss
LAMBDA_BC Loss
LAMBDA_PDE Loss
STAGE Loss


Adaptive Loss Weights Over Training
Weight value
Epoch
λ_ic
λ_bc
λ_pde
Curriculum Stage Progression
Curriculum Stage
Epoch


AC-PINN - Allen-Cahn Clean Dense
t
x
u(x,t)



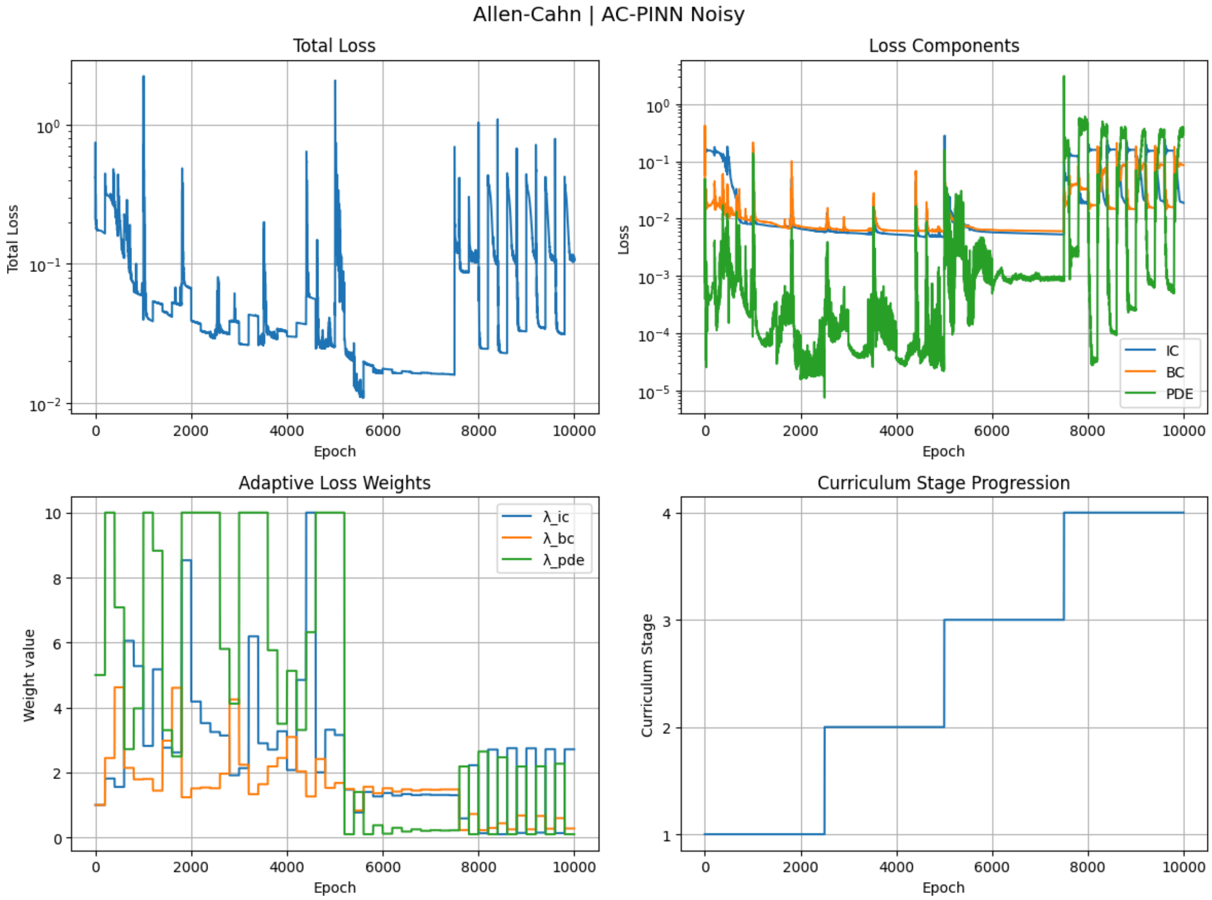

```
Saved: ../results/allen_cahn/ac_clean_history.npy
```

# Section 6 - AC-PINN, Noisy Sparse

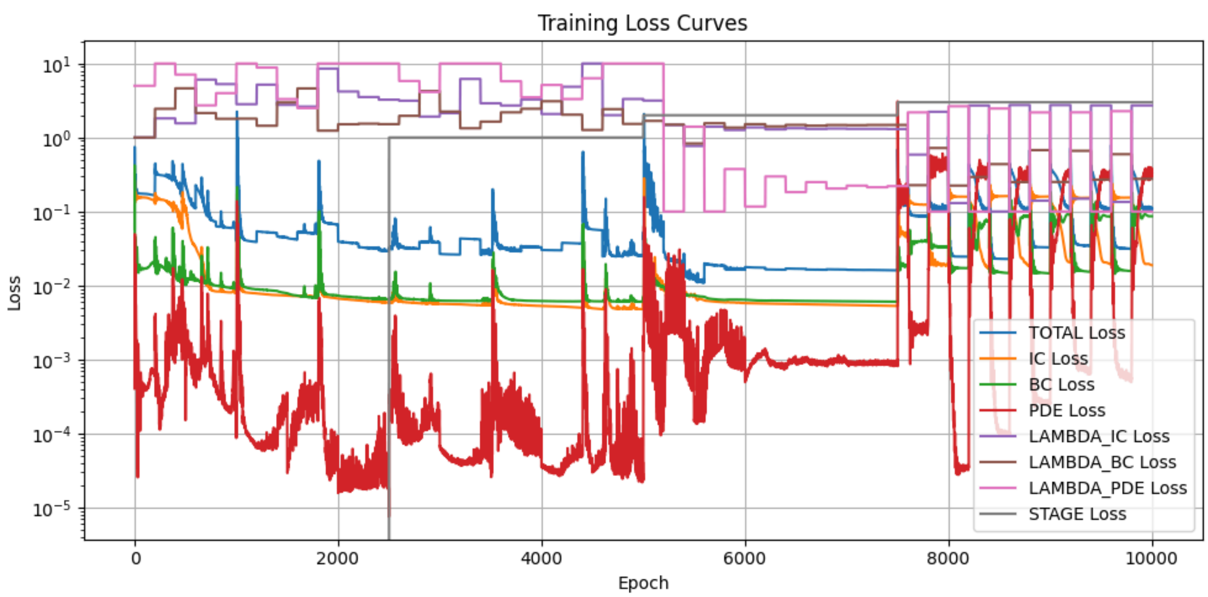
This is where AC-PINN should demonstrate its biggest advantage.

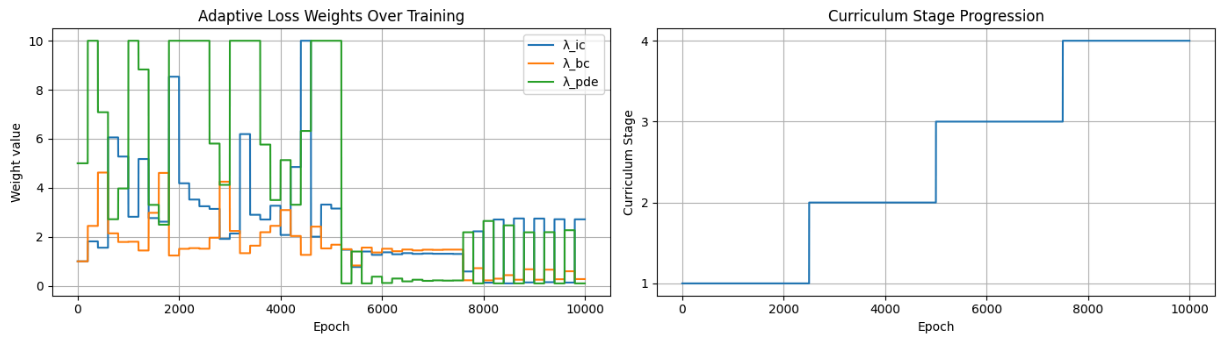
In [7]:
```
ac_noisy = ACPINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS,
                        weight_strategy='both',
                        N_pool=30000, resample_every=500)
h_an = ac_noisy.fit(data_noisy_sparse, epochs=EPOCHS, print_every=1000, label='Alle
save_training_plots(h_an, FIGURES+'acpinn_noisy_training.png', 'Allen-Cahn | AC-PIN
ac_noisy.plot_loss_history(h_an)
ac_noisy.plot_weight_history(h_an)
ac_noisy.plot_solution(title='AC-PINN - Allen-Cahn Noisy Sparse')
save_history(h_an, RESULTS+'ac_noisy_history.npy')
```

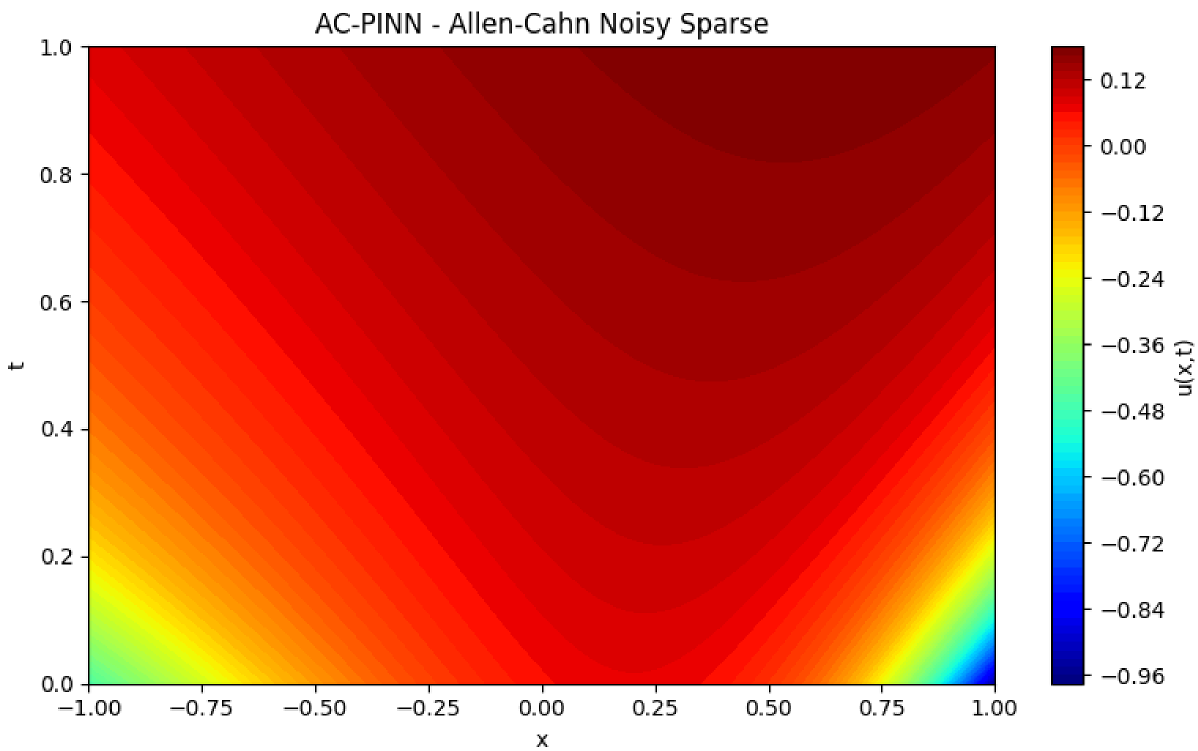
```
[Allen-Cahn | AC-PINN Noisy | Epoch     0/10000] Stage 1/4 | Total: 0.211765 | IC: 
0.150837 | BC: 0.058899 | PDE: 0.000406 | λ=(1.00,1.00,5.00)
[Allen-Cahn | AC-PINN Noisy | Epoch  1000/10000] Stage 1/4 | Total: 0.039671 | IC: 
0.007976 | BC: 0.009067 | PDE: 0.000087 | λ=(2.82,1.80,10.00)
[Allen-Cahn | AC-PINN Noisy | Epoch  2000/10000] Stage 1/4 | Total: 0.039165 | IC: 
0.006673 | BC: 0.007338 | PDE: 0.000016 | λ=(4.18,1.51,10.00)
[Allen-Cahn | AC-PINN Noisy | Epoch  3000/10000] Stage 2/4 | Total: 0.026770 | IC: 
0.005654 | BC: 0.006264 | PDE: 0.000068 | λ=(2.13,2.24,10.00)
[Allen-Cahn | AC-PINN Noisy | Epoch  4000/10000] Stage 2/4 | Total: 0.030289 | IC: 
0.005303 | BC: 0.006176 | PDE: 0.000036 | λ=(2.08,3.09,5.13)
[Allen-Cahn | AC-PINN Noisy | Epoch  5000/10000] Stage 2/4 | Total: 0.025561 | IC: 
0.004796 | BC: 0.006078 | PDE: 0.000023 | λ=(3.15,1.68,10.00)
[Allen-Cahn | AC-PINN Noisy | Epoch  6000/10000] Stage 3/4 | Total: 0.017738 | IC: 
0.005808 | BC: 0.006425 | PDE: 0.000495 | λ=(1.37,1.51,0.12)
[Allen-Cahn | AC-PINN Noisy | Epoch  7000/10000] Stage 3/4 | Total: 0.016336 | IC: 
0.005461 | BC: 0.006119 | PDE: 0.000933 | λ=(1.31,1.47,0.22)
[Allen-Cahn | AC-PINN Noisy | Epoch  8000/10000] Stage 4/4 | Total: 1.036718 | IC: 
0.019074 | BC: 0.032861 | PDE: 0.388097 | λ=(0.13,0.22,2.65)
[Allen-Cahn | AC-PINN Noisy | Epoch  9000/10000] Stage 4/4 | Total: 0.441073 | IC: 
0.159359 | BC: 0.014571 | PDE: 0.000252 | λ=(2.74,0.25,0.10)
AC-PINN training complete in 158.41s
```




```
Saved: ../figures/allen_cahn/acpinn_noisy_training.png
Saved: ../figures/allen_cahn/acpinn_noisy_training.csv
```





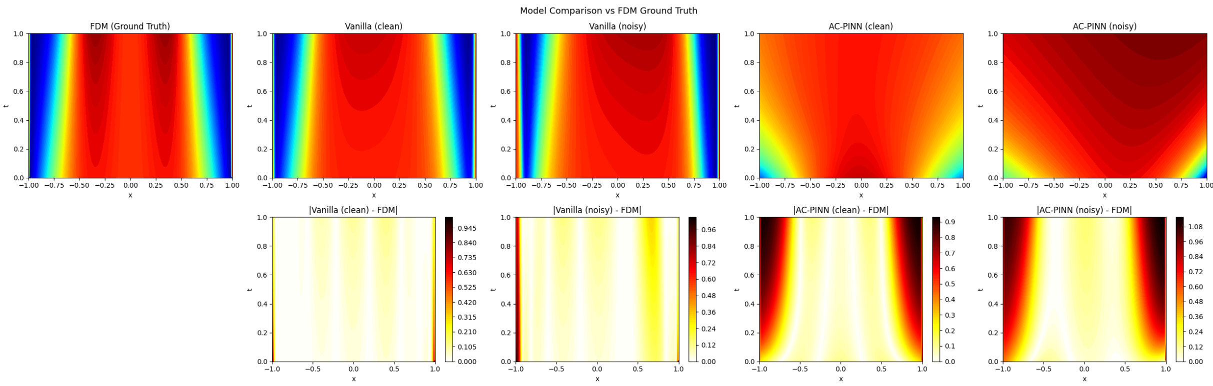

```
Saved: ../results/allen_cahn/ac_noisy_history.npy
```

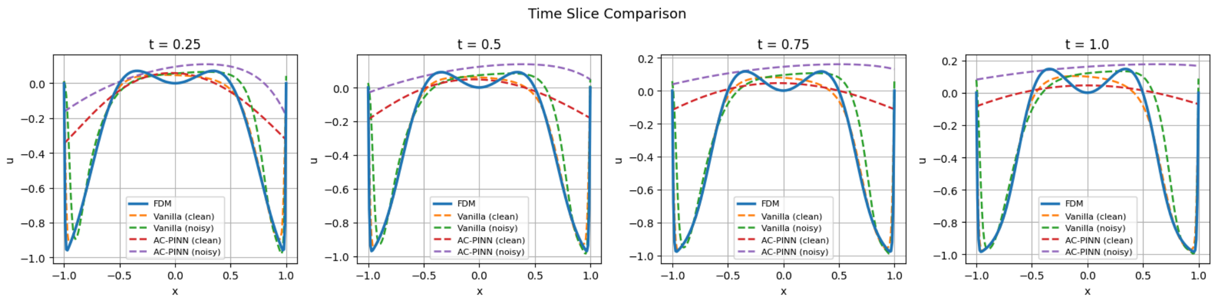
# Section 7 - Benchmark vs FDM

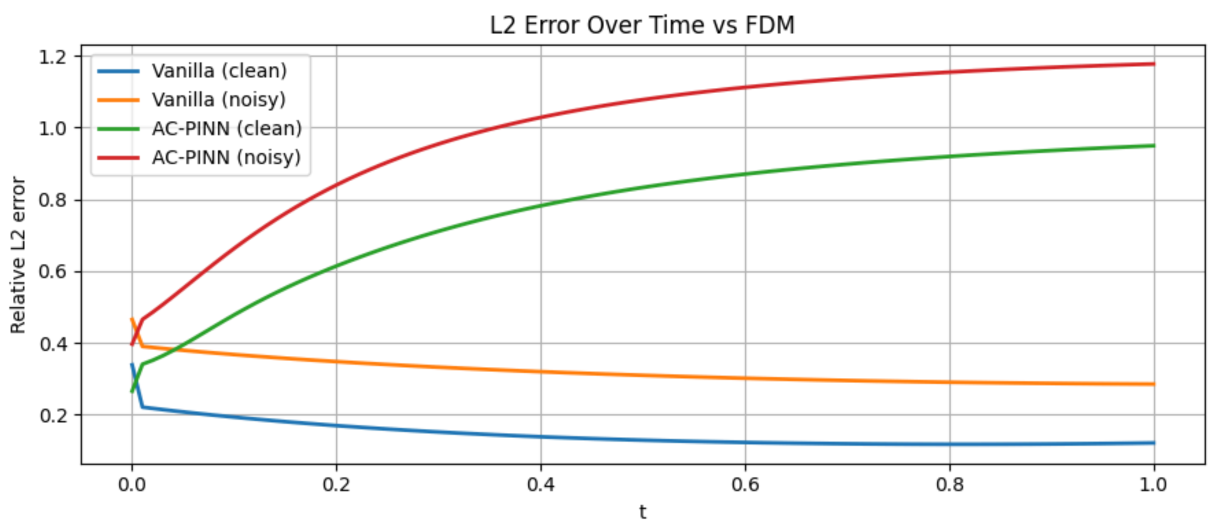
In [8]:

```
bench = Benchmark(fdm, nx=200, nt=100)
bench.add('Vanilla (clean)', vanilla_clean)
bench.add('Vanilla (noisy)', vanilla_noisy)
bench.add('AC-PINN (clean)', ac_clean)
bench.add('AC-PINN (noisy)', ac_noisy)
bench.run()

metrics = bench.compare_metrics()
bench.plot_comparison(save_path=FIGURES+'comparison.png')
bench.plot_time_slices(save_path=FIGURES+'time_slices.png')
bench.plot_error_over_time(save_path=FIGURES+'error_over_time.png')

save_metrics(metrics, RESULTS+'benchmark_metrics.npy')
print('Allen-Cahn experiments complete.')
```

```
=================================================================
  Model                          Rel L2     Max Err        MAE       RMSE
=================================================================
  Vanilla (clean)              0.143559   1.017695   0.035595   0.062932
  Vanilla (noisy)              0.316406   1.047791   0.077888   0.138703
  AC-PINN (clean)              0.812069   0.922218   0.229725   0.355987
  AC-PINN (noisy)              1.040699   1.153148   0.309040   0.456213
=================================================================
```




```
Saved: ../results/allen_cahn/benchmark_metrics.npy
Allen-Cahn experiments complete.
```

# Section 8 - Noise Level Study

In [9]:
```python
noise_results = {}
for eps in [0.05, 0.1, 0.2]:
    print(f'\n--- ε={eps} ---')
    d = gen.generate(N_ic=50, N_bc=50, N_f=5000, noise_eps=eps)
    v = PINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS, lambda_pde=10.0)
```

```
    v.fit(d, epochs=5000, print_every=2500)
    a = ACPINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS, weight_strategy
    a.fit(d, epochs=5000, print_every=2500)
    b = Benchmark(fdm).add(f'Vanilla ε={eps}', v).add(f'AC-PINN ε={eps}', a)
    b.run()
    noise_results[eps] = b.compare_metrics()
save_metrics(noise_results, RESULTS+'noise_study_metrics.npy')
print('Allen-Cahn noise study complete.')
```

```
--- ε=0.05 ---
[Epoch     0/5000] Total: 0.591497 | IC: 0.160854 | BC: 0.155205 | PDE: 0.027544
[Epoch  2500/5000] Total: 0.124157 | IC: 0.112330 | BC: 0.009061 | PDE: 0.000277
Training complete in 86.70s
[Epoch     0/5000] Stage 1/4 | Total: 0.127733 | IC: 0.119624 | BC: 0.007746 | PDE:
0.000073 | λ=(1.00,1.00,5.00)
[Epoch  2500/5000] Stage 2/4 | Total: 0.015569 | IC: 0.002428 | BC: 0.002987 | PDE:
0.000005 | λ=(3.77,2.14,3.74)
AC-PINN training complete in 93.99s
=================================================================
  Model                          Rel L2    Max Err        MAE       RMSE
=================================================================
  Vanilla ε=0.05               0.549171   1.018321   0.132007   0.240741
  AC-PINN ε=0.05               0.848223   0.958067   0.241095   0.371837
=================================================================

--- ε=0.1 ---
[Epoch     0/5000] Total: 0.811225 | IC: 0.213705 | BC: 0.222257 | PDE: 0.037526
[Epoch  2500/5000] Total: 0.159354 | IC: 0.137152 | BC: 0.018529 | PDE: 0.000367
Training complete in 85.38s
[Epoch     0/5000] Stage 1/4 | Total: 0.185915 | IC: 0.159454 | BC: 0.025178 | PDE:
0.000256 | λ=(1.00,1.00,5.00)
[Epoch  2500/5000] Stage 2/4 | Total: 0.045237 | IC: 0.005578 | BC: 0.012597 | PDE:
0.000016 | λ=(4.60,1.54,7.43)
AC-PINN training complete in 88.18s
=================================================================
  Model                          Rel L2    Max Err        MAE       RMSE
=================================================================
  Vanilla ε=0.1                0.794291   0.974870   0.248946   0.348194
  AC-PINN ε=0.1                0.979001   0.990372   0.288530   0.429166
=================================================================

--- ε=0.2 ---
[Epoch     0/5000] Total: 0.268482 | IC: 0.125368 | BC: 0.076212 | PDE: 0.006690
[Epoch  2500/5000] Total: 0.178392 | IC: 0.086120 | BC: 0.069230 | PDE: 0.002304
Training complete in 81.85s
[Epoch     0/5000] Stage 1/4 | Total: 0.193024 | IC: 0.121321 | BC: 0.070732 | PDE:
0.000194 | λ=(1.00,1.00,5.00)
[Epoch  2500/5000] Stage 2/4 | Total: 0.233589 | IC: 0.042495 | BC: 0.063939 | PDE:
0.000048 | λ=(2.78,1.80,10.00)
```

```
AC-PINN training complete in 88.19s
=================================================================
  Model                         Rel L2    Max Err        MAE       RMSE
=================================================================
  Vanilla ε=0.2               0.459016   1.069122   0.109629   0.201219
  AC-PINN ε=0.2               0.770576   1.206568   0.217735   0.337798
=================================================================
Saved: ../results/allen_cahn/noise_study_metrics.npy
Allen-Cahn noise study complete.
```